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An Intelligent Multi-Criteria Decision Approach for Selecting the Optimal Operating System for Educational Environments

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Abstract: The selection of an appropriate operating system (OS) in educational environments is a critical decision that impacts the functionality, user experience, and overall efficiency. Financial factors, along with the availability and functionality of tools used on these systems, play a crucial role in this selection. Furthermore, the OS affects the user experience, security, and adaptability to learning. Previous research in this area is scarce, and this paper contributes to a better understanding of the OS selection process in education. This paper proposes a novel multi-criteria decision-making approach to evaluate and select the most suitable OS for educational institutions. The methodology integrates the PIPRECIA method for assessing weighted criteria, alongside utility analysis (NWA), enabling a balanced decision-making process between institutional management and IT experts. The evaluation considered the following criteria: performance, cost, security, usability, implementation, support, and documentation. The study compared three popular operating systems: Microsoft Windows 11, GNU Linux Ubuntu 22.04 LTS, and Apple macOS 12 using the proposed approach. The results, based on the integrated evaluation of all criteria, indicate that GNU Linux Ubuntu 22.04 LTS (0.562) ranked highest, followed by Microsoft Windows 11 (0.553) and Apple macOS 12 (0.543). This paper emphasizes the importance of objective, group-based decision-making in selecting an OS for education, providing practical insights and guidelines for effective technology integration in academic settings.

Keywords: education; multi-criteria decision making; net worth analysis; operating system; PIPRECIA method

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1. Introduction

In today's digital age, technology has become an indispensable part of the educational process [1,2]. With an increasing number of digital tools and resources available, the choice of the appropriate operating system has become a crucial factor for efficient work in education. The OS is not only the foundation for computer functioning, but also plays a crucial role in adapting to and supporting [3] a variety of tools and resources [4] used in education [5].

One of the key advantages of the PIPRECIA method compared to other multi-criteria decision-making methods, such as AHP or TOPSIS, is its simplicity and speed of implementation. While AHP requires complex hierarchical analysis and involves multiple pairwise comparisons between criteria, which can increase subjectivity and demand significant time and resources, PIPRECIA offers a more linear and intuitive decision-making process. This feature is particularly useful in educational contexts where decisions are often made under time constraints and with limited resources. Additionally, PIPRECIA reduces the potential for errors in judgment through a simplified weighting procedure, which is a critical advantage in the educational sector with its specific demands for transparency and efficiency.

The financial aspect of choosing an OS has a direct impact on the budgets of institutions and individuals in education [6]. Different OSs have different licensing costs, and they also vary in their hardware resource requirements [7]. For example, an open-source OS like Linux can provide significant savings as they are free to use [8] and often require fewer resources than commercial alternatives such as Windows or macOS. These savings can be essential for institutions with limited budgets [9], allowing them to invest funds in other educational needs.

In addition to the financial aspect, the choice of OS also affects the availability and functionality of various tools and applications used in education [10]. While some tools may have support for all popular OSs, others may only be optimized for a specific OS. For example, certain video editing or design software may be available only for certain platforms, which can limit the options for students or teachers if the OS they use is not supported.

The impact of the choice of OS on the educational process can be profound and wide-ranging. The OS not only provides a basic platform for running software [11], but also influences the user experience, system reliability, security, and accessibility. For example, an OS with an intuitive user interface can make it easier for students and teachers to quickly learn and use various tools and resources. Additionally, the stability and security of the OS are crucial for maintaining data integrity and protecting privacy in education, especially in environments where sensitive information is often exchanged.

Furthermore, the OS can also have implications [12] for the teaching and learning process itself. For example, platforms that support sharing and collaboration among users can encourage interactive and teamwork-based teaching [13], while specific OS features such as digital whiteboards or content customization functions can facilitate the personalized learning and adaptation [14] of educational experiences [15] to different students [16].

Choosing an efficient working environment is crucial but also very difficult to make. In practice, the path of least resistance is usually followed, leaning toward existing systems, which is not always the right choice. Proper choices can be made based on multiple criteria. One technology often applied and is very important is the computer OS. An OS is software that acts as an interface between the user of a computer and the computer hardware and also manages and coordinates the use of hardware among various applications and users [17]. Commonly used OSs include MS Windows, Macintosh (MacOS), and GNU/Linux.

The rest of this paper is structured as follows. Section 2 provides a review of previous research, highlighting the existing gaps and the need for a systematic decision-making approach in operating system selection. Section 3 introduces the model for multi-criteria decision-making in education, detailing the methodologies employed including PIPRECIA and net worth analysis (NWA). This section describes the computational procedures of the PIPRECIA method, outlining the steps for determining criteria weights and their significance in the decision-making process. Section 4 presents the research findings including the application of the methods to evaluate and rank operating systems. Section 5 presents a sensitivity analysis that assesses how variations in the weight of all criteria affect the final ranking of operating systems. Section 6 discusses the key results, emphasizing the implications, advantages, and limitations of the proposed approach. Finally, Section 7 concludes the paper by summarizing the contributions and providing recommendations for future research.

2. Previous Research

Previous research on the topic of OS selection in education is scarce, and the importance of this issue is increasingly highlighted in today's digital age. Researchers have addressed various aspects of this problem, but there is a clear need for a comprehensive review of methods and approaches.

It is necessary to consider a large number of criteria when choosing an OS such as budget, hardware, and implementation. This process requires the thorough testing of all offered OSs on existing hardware, along with the use of appropriate application programs. However, this procedure is often slow, demanding, and costly, so a long-term selection method needs to be found, in consultation with IT experts.

There is a considerable amount of research dealing with OSs in education [18–22], but from a completely different perspective on the issue. Only two groups of authors have dealt with the decision theory of OS implementation. The first study [23] dealt with evaluating OSs for all applications and used the AHP method, and in the second study [24], an identical study was conducted only with the application of the AHP method, which is often used but that we believe needs to be upgraded and some newer methods applied, as in our case, the Pivot Pairwise Relatives Criteria Importance Assessment (PIPRECIA).

In the context of decision-making in education, where technology is rapidly changing, the challenge is to develop a sufficiently good model for selecting the optimal solution. The aim of the paper was to propose a model that can be used in education to select computer equipment and software, especially an OS.

The proposed method in this paper combines the knowledge of both sides: decision-makers and IT experts, in such a way that each participant performs their part of the work. In the end, by applying this method, the best proposed solution tailored to the capabilities and needs of the institution is obtained.

Practically, there has been almost no similar research conducted on OSs using any of the multi-criteria decision-making methods, especially not using the PIPRECIA method.

Although relatively recently proposed, the PIPRECIA method has been widely used to define the weights of various types of criteria. Some of the applications of this method include evaluating the quality of websites in the hotel industry [25]; solving problems related to the selection of the optimal mining method [26]; selecting the optimal quality control manager [27]; and evaluating the quality of materials intended for distance learning [28]. The presented studies confirm that the PIPRECIA method is useful and will also enable relevant results to be obtained in the field of choosing the appropriate OS for educational institutions.

Later, modified versions of the PIPRECIA method appeared in the form of phase PIPRECIA for analyzing information systems in the warehouse system [29] as well as evaluating the criteria for implementing high-performance computing in Danube region countries [30]. There has also been the emergence of PIPRECIA S (simplified) [31].

Although previous research is rare, this paper represents a step forward in better understanding the process of OS selection in education. We will now carefully consider the methodology used in this research and its implications for the educational process.

Methods such as AHP and TOPSIS have been extensively used in multi-criteria decision-making, but exhibit certain limitations when applied in the context of educational environments. AHP, for instance, involves complex hierarchical modeling and numerous pairwise comparisons, which can introduce subjectivity and increase the time required for decision-making. Similarly, TOPSIS relies heavily on the determination of ideal and negative-ideal solutions, which may not always align with the specific needs and constraints of educational institutions. In contrast, the PIPRECIA method offers a more straightforward and intuitive process for determining the relative importance of criteria, minimizing subjectivity and computational complexity. This simplicity makes it particularly well-suited for educational settings where decisions often need to be made efficiently and transparently. Furthermore, PIPRECIA's linear evaluation framework ensures consistency and reduces the likelihood of errors during the weighting process, which is crucial for maintaining the reliability of decision-making in resource-constrained educational environments.

3. Materials and Methods

3.1. Multi-Criteria Methods

Multi-criteria decision-making methods play a crucial role in the decision-making process, providing a structured approach to ranking options according to multiple criteria [32]. This research employed various methods to analyze OSs in the context of education.

Some of the commonly used methods include the analytic hierarchy process (AHP) [33], which establishes a hierarchical structure of criteria and alternatives, assigning weights to criteria and comparing options pairwise against each criterion. TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) [34,35] is based on ideal and negative ideal solutions, aiding in determining the ranking of alternatives based on their similarity to the ideal solution. ELECTRE (Elimination Et Choice Traduisant la Réalité) [36,37] utilizes a preference matrix to rank alternatives based on preferences and non-preferences. Vikor [38,39] identified a compromise solution closest to the ideal solution considering optimality and stability. The weighted sum model [40] combines multiple criteria into a single overall score for each alternative.

These methods enable a structured and systematic approach to decision-making, allowing decision-makers in education to consider various aspects and priorities when selecting an OS.

This research places special emphasis on the application of multi-criteria methods in evaluating OSs in education. Through methods such as PIPRECIA (Subjective-Partial Pairwise Comparison Method) [41], operating systems are evaluated based on given criteria, taking into account subjective and partial pairwise comparisons. The result of this research is a transparent summation of the value of OSs, considering the fulfillment of requirements for each criterion.

The idea of this research is to enable school managers to make precise and reliable decisions regarding the choice of an operating system (OS). To achieve this, two groups of decision-makers were engaged. The first group consisted of three educational managers, responsible for defining the requirements and criteria based on the specific needs of educational institutions. Their expertise included school administration, curriculum development, and IT infrastructure planning in education. The second group comprised two IT experts with extensive experience in system administration and cybersecurity who provided expert assessments based on a defined scale. By combining the insights of these two groups, this research ensured a balanced approach that addressed both the technical and operational aspects of OS evaluation. This collaborative effort contributes to the development of a practically applicable model for evaluating OSs in education, ensuring its relevance and usability in real-world scenarios.

3.2. PIPRECIA Method

The PIPRECIA method allows for the evaluation of criteria without the prior sorting of criteria by importance [28,42]. This method is used to determine the weights of criteria when the criteria are both quantitative and qualitative [43]. The steps are simpler compared with the AHP method [33], and the weights of the criteria do not depend on the subjective reasoning of decision-makers, but only on the comparison of importance between criteria. Detailed steps of this method will be presented in the second part of this paper. To date, there have been no studies that have applied this method to determine the weights of criteria in the decision-making process for selecting an OS. This gap is the reason for conducting this research using PIPRECIA [41].

The first phase of ranking alternatives involves determining the significance or weights of the criteria. There are many methods for determining the weights of criteria, and in this work, the PIPRECIA method was used, which is an enhanced version of the SWARA method [44] applied in the decision-making group.

3.2.1. Computational Procedure

The computational procedure of the PIPRECIA can be expressed as follows [41]:

Step 1. Determine the relevant evaluation criteria and sort them in descending order, based on their expected significance.

Step 2. Starting from the second criterion, determine the relative importance s_j based on Equation (1).

$$s_j = \begin{cases} > 1,\ C_j > C_1 \\ = 1,\ C_j = C_1 \\ < 1,\ C_j < C_1 \end{cases} \tag{1}$$

Initially, the criterion C_1 is defined as the reference for comparison. Moving on to the second criterion, each criterion C_j is assigned a relative importance value s_j, as defined by Equation (1). This means that each criterion C_j is compared to the reference criterion C_1. If criterion C_j is considered more important than C_1, it is given a value s_j greater than 1. Conversely, if C_j is deemed less important than C_1, it is assigned a value less than 1. When both criteria C_1 and C_j are equally important, they are both assigned an importance value of 1.

Step 3. The value of the coefficient k_j is calculated based on Equation (2).

$$k_j = \begin{cases} 1,\ j = 1 \\ 2 - s_j,\ j > 1 \end{cases} \tag{2}$$

Step 4. The value of the coefficient q_j is calculated based on Equation (3).

$$q_j = \begin{cases} 1,\ j = 1 \\ \frac{q_{j-1}}{k_j},\ j > 1 \end{cases} \tag{3}$$

Step 5. Calculate the relative weight w_j of the criteria. Based on Equation (4), the relative weight of criteria is calculated, w_j, where $0 \le w_j \le 1$ and $\sum_{k=1}^{n} w_k = 1$.

$$w_j = \frac{q_j}{\sum_{k=1}^{n} q_k} \tag{4}$$

3.2.2. Implementing in Decision-Making Environment

Many real decision-making problems require the involvement of several, or even more, decision-makers. In such cases, the resulting weights that represent the attitude of the group made up of K decision-makers can be carried out using Equations (5) and (6).

$$w_j^* = \left(\prod_{r=1}^{R} w_j^{nr} \right)^{1/R} \tag{5}$$

$$w_j = \frac{w_j^*}{\sum_{j=1}^{n} w_j^*} \tag{6}$$

where w_j^* denotes the geometric mean of the weights of criterion j obtained by surveying respondents.

3.2.3. Net Worth Analysis (NWA)

Multi-criteria decision-making (MCDM) is the process of decision-making where multiple criteria or factors are taken into account to select the best possible choice among different alternatives [45–48]. This method is often used in situations where it is necessary to consider multiple aspects or goals, and one solution may not adequately satisfy all of those goals. MCDM can be a useful tool in many areas including business decisions, resource management, product or service selection, and project planning as well as in education and public policy. The main goal of MCDM is the identification and ranking of alternatives according to predefined criteria to make an informed decision that best meets the needs or goals of an organization or individual.

Multi-criteria decision-making (MCDM) refers to decision-making situations where there are multiple, often conflicting criteria. Multi-attribute utility theory represents a very specific part of multi-criteria decision-making [36,49].

The decision between different alternatives is often made based on one's own experiences and intuition. Net worth analysis (NWA) is a suitable tool for the systematic evaluation and classification of alternatives and the identification of the most suitable one [50]. The evaluation process is simple and transparent. Net worth analysis is a method for determining the best solution. To this end, criteria for evaluating solutions must first be defined.

According to [50], net worth analysis (NWA) involves analyzing a set of complex alternatives to arrange the elements of that set depending on the decision-maker, considering the goal expressed by some multi-dimensional system of evaluation criteria. This arrangement is shown by indicating the overall values of the alternatives. Net worth analysis assumes that when the overall value is greater, the individual values are higher.

The values of the OS considering significance factors can be calculated according to Equation (7).

$$N = \sum_{j=1}^{9} g_j \cdot n_{ij} \tag{7}$$

The PIPRECIA and NWA methods were applied in their standard form; this study acknowledges the potential for further methodological enhancements to address more complex or uncertain decision-making scenarios. For example, the incorporation of fuzzy logic could provide additional robustness by allowing for the inclusion of imprecise or subjective data, which is often encountered in real-world educational contexts. Fuzzy extensions of the PIPRECIA method, such as fuzzy PIPRECIA or PIPRECIA-S with fuzzy weights, could enhance the adaptability of the model when dealing with ambiguity in expert assessments or varying priorities among stakeholders.

However, the choice to employ the standard PIPRECIA and NWA methods in this study was deliberate, as these methods offer a balance between simplicity, transparency, and effectiveness, key attributes for decision-making in resource-constrained educational environments. Future work could explore integrating fuzzy logic or similar adaptations to expand the applicability of this approach to more complex and uncertain scenarios such as large-scale educational reforms or technology implementations with high variability in stakeholder input.

3.2.4. Expert Involvement and Validation

The involvement of experts in this study was critical for ensuring the validity and practical relevance of the proposed model. The expert panel consisted of three educational managers and two IT professionals. The educational managers, with an average of over 10 years of experience in school administration, curriculum development, and educational resource planning, provided insights into the operational and strategic priorities of educational institutions. The IT professionals, each with over 15 years of experience in system administration and cybersecurity, contributed their technical expertise to evaluate the feasibility and reliability of the selected operating systems.

This combination of perspectives allowed for a comprehensive and balanced approach to decision-making, integrating practical and technical considerations. Furthermore, the use of group decision-making minimized individual biases, enhancing the overall reliability of the results. The experts' feedback also played a vital role in validating the final rankings, ensuring their alignment with real-world requirements in educational settings.

4. Results

4.1. Determining the Criteria and Criteria Significance

In order to determine the appropriate OS for educational institutions, it is first necessary to define the criteria and determine their significance factors.

Criteria for evaluating the OS are very rare, especially when it comes to education, which brings its own unique challenges. In [23], seven criteria were used, deemed relevant for evaluating the OS. Their research focused on the steps to follow when choosing an OS, applicable across various sectors. For a more precise analysis dedicated to a specific sector such as education, this study proposed a methodology that can be applied in education using available resources. Therefore, some criteria were modified and tailored to the needs arising in educational institutions.

In [24], the AHP method was used, and to calculate the group decisions, the average values among decision-makers were utilized. The authors of this research are employed in education and thus frequently encounter decision-making regarding equipment and software used in the teaching process. What complicates making the right decision is the difference in understanding the challenges between management (DM) and the IT personnel responsible for implementing the decisions. This method mathematically makes decisions based on the independent opinions of each decision-maker as well as the IT personnel, hence considered to be valid and represent the best decision.

Based on the previous research above-mentioned and the authors' experience, the following criteria for evaluation were selected: Performance (C1), Cost (C2), Security (C3), Usability (C4), Implementation (C5), Support (C6), and Documentation (C7).

Performance (C1): When evaluating performance, we refer to hardware and how a particular OS behaves on older or lower-end computer configurations, which are quite common in education. Here lies a challenge, as Apple comes with its own hardware, and if the school does not possess it, the costs are significantly higher, or if it does, then the challenges are on the side of the other two OSs in our analysis.

Cost (C2): This criterion is significant for educational institutions with small budgets always seeking ways to save. This criterion also poses challenges similar to performance in terms of acquiring new equipment, depending on the institution's situation or the chosen OS, which can vary greatly.

Security (C3): In today's age of general digitization, this is a very important criterion, especially when dealing with younger individuals as is the case of education.

Usability (C4): Refers to the usability in relation to the needs of educational programs taught in schools.

Implementation (C5): In terms of hardware, educational programs, the literature available to teachers and students as well as the technical abilities of those responsible for maintenance within the institution.

Support (C6): OS support can vary, either institutionally where the manufacturer provides support, or online in the form of a developed community supporting users. This criterion is very important, especially when transitioning to a new OS or in the case of any issues that users are unable to solve on their own.

Documentation (C7): Documentation or the literature available for learning, both for teaching staff and for the students themselves. Besides this main task, documentation also provides a form of support through which users can solve problems they encounter in their work.

4.2. Computing the Relative Importance (s_j)

The decision-making process in an educational institution often requires analysis from various perspectives and expertise. The management of the educational institution, consisting of three members, plays a crucial role in setting the criteria and priorities for decision-making, while the other team, composed of experts in the field of information technology, contributes expertise on the technical aspects of the decisions.

This dynamic allows for the integration of different perspectives into the decision-making process. The internal computer science professor brings a deep understanding of the specific needs of the educational institution, while the external expert can bring fresh ideas from a broader industrial context.

The combination of internal and external knowledge enables a holistic approach to problem-solving and process optimization in the educational environment. The evaluation results of each individual DM are an important phase of method application and objective criterion setting (Table 1). The priority scale in this method ranges from 0.60 to 1.80, where 1.00 represents equal importance.

Table 1. Results of defined criteria for each DM.

Criterion	DM1 (s_j)	DM2 (s_j)	DM3 (s_j)
C1			
C2	1.30	1.50	1.10
C3	1.00	1.25	1.00
C4	0.90	1.00	1.00
C5	0.85	0.95	0.90
C6	0.65	0.90	0.60
C7	0.60	0.60	0.60

4.3. Computing the Coefficient (k_j)

Table 2 shows the results of the defined coefficient for each DM.

Table 2. Results of the defined coefficients for each DM.

Criterion	DM1 (k_j)	DM2 (k_j)	DM3 (k_j)
C1	1.00	1.00	1.00
C2	0.70	0.50	0.90
C3	1.00	0.75	1.00
C4	1.10	1.00	1.00
C5	1.15	1.05	1.10
C6	1.35	1.10	1.40
C7	1.40	1.40	1.40

4.4. Computing the Recalculated Values (qj)

The following table (Table 3) displays the recalculated values for each individual DM.

Table 3. Results of the recalculated values for each DM.

Criterion	DM1 (q_j)	DM2 (q_j)	DM3 (q_j)
C1	1.00	1.00	1.00
C2	1.43	2.00	1.11
C3	1.43	2.67	1.11
C4	1.30	2.67	1.11
C5	1.13	2.54	1.01
C6	0.84	2.31	0.72
C7	0.60	1.65	0.52

4.5. Computing the Relative Weights of Evaluation Criteria

The following table (Table 4) displays the relative weight parameters for each individual DM as well as their average values.

Table 4. Results of the defined relative weight criteria for each DM and their average value.

Criterion	DM1 (w_j)	DM2 (w_j)	DM3 (w_j)
C1	0.1295	0.0674	0.1520
C2	0.1851	0.1349	0.1689
C3	0.1851	0.1798	0.1689
C4	0.1682	0.1798	0.1689
C5	0.1463	0.1712	0.1535
C6	0.1084	0.1557	0.1096
C7	0.0774	0.1112	0.0783

4.6. Computing the Total Value of OS

After defining the relative weight criteria by the decision-makers, the final value of each OS was determined using the utility analysis method. Ratings from 1 to 5 were used to evaluate the OS, with the experts utilizing the Likert scale [51] for assessment.

Defining a scale from 1 to 5 is a crucial step in the decision-making process, allowing for clear differentiation between different options and facilitating communication within teams. Evaluation criteria need to be clear and measurable to avoid subjectivity and ensure consistency in assessment.

The Likert scale, which enables respondents to express their opinions continuously, is often used to measure attitudes and opinions in research studies and surveys [52]. This scale facilitates the collection of quantitative data on the respondents' attitudes and opinions, which is particularly useful in educational research and course evaluations.

In this case, the expert plays a crucial role in evaluating an OS, relying on their knowledge and experience to assess the current situation within the institution and propose the best solutions.

Table 5 shows the results of the defined criteria cross-referenced with the ratings of IT experts.

Table 5. Results of defined criteria cross-referenced with the IT expert ratings.

Criterion	Significance Factor	Microsoft Windows 11		GNU Linux Ubuntu 22.04 LTS		Apple Mac OS X 12	
		Rating	Points	Rating	Points	Rating	Points
C_1	0.1099	3	0.330	5	0.549	4	0.440
C_2	0.1616	3	0.485	5	0.808	2	0.323
C_3	0.1778	3	0.533	5	0.889	4	0.711
C_4	0.1722	4	0.689	3	0.517	5	0.861
C_5	0.1567	5	0.783	4	0.627	4	0.627
C_6	0.1228	5	0.614	3	0.368	4	0.491
C_7	0.0877	5	0.438	2	0.175	4	0.351
Score	0.9886	4	0.553	4	0.562	4	0.543
		2		1		3	

5. Sensitivity Analysis

To assess the robustness of the proposed decision-making model, a comprehensive sensitivity analysis was conducted. This analysis examined how variations in the weights of each of the seven criteria affected the final rankings of the three evaluated operating systems: GNU Linux Ubuntu 22.04 LTS, Microsoft Windows 11, and Apple macOS 12.

For each criterion, the weight was incrementally adjusted by ±10%, ±20%, and ±30%, while the weights of the other criteria were held constant. The adjusted weights and their impact on the final scores and rankings are summarized in Figures 1–7.

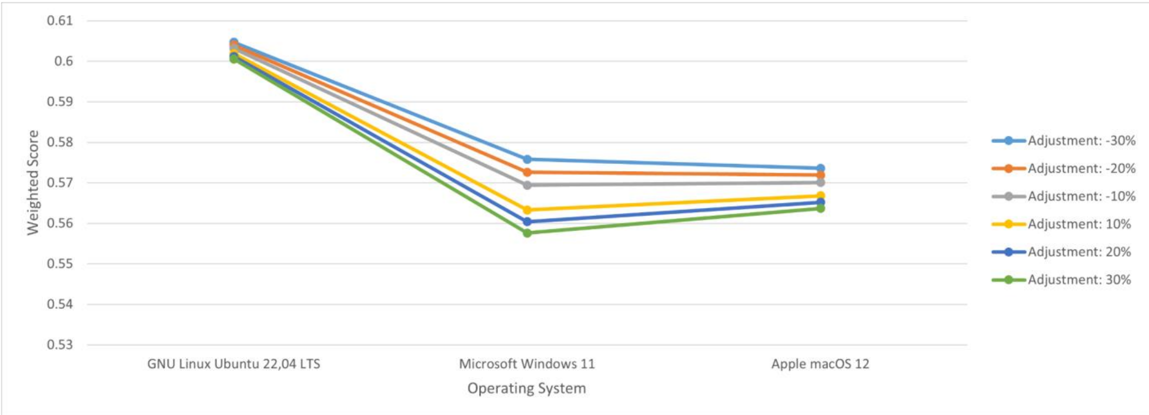

Figure 1. Sensitivity analysis for performance.

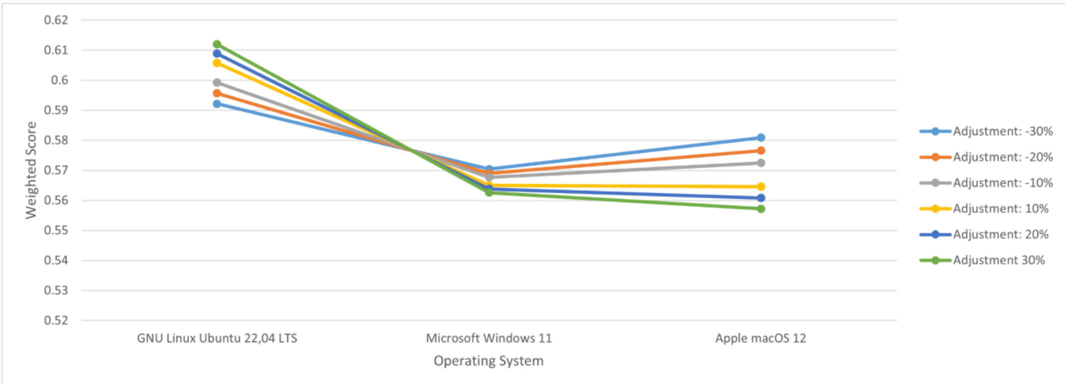

Figure 2. Sensitivity analysis for cost.

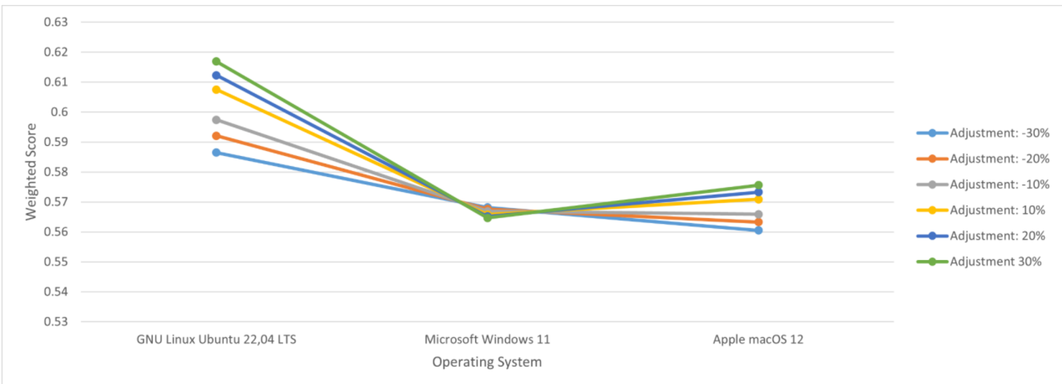

Figure 3. Sensitivity analysis for security.

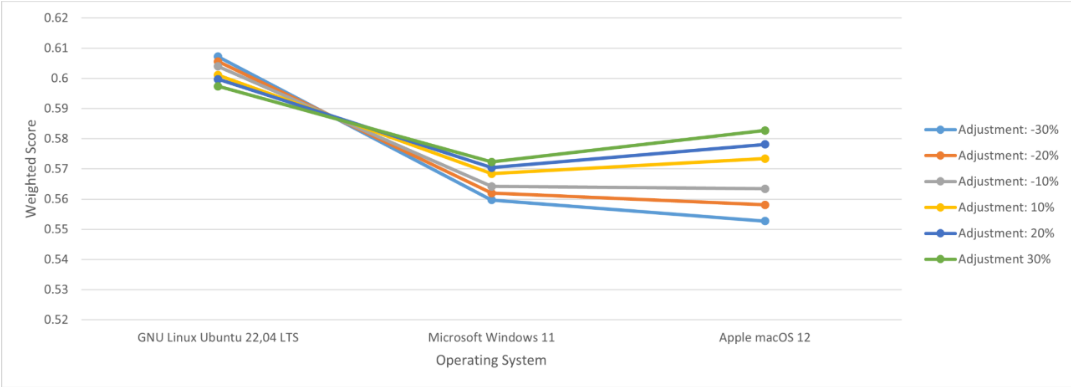

Figure 4. Sensitivity analysis for usability.

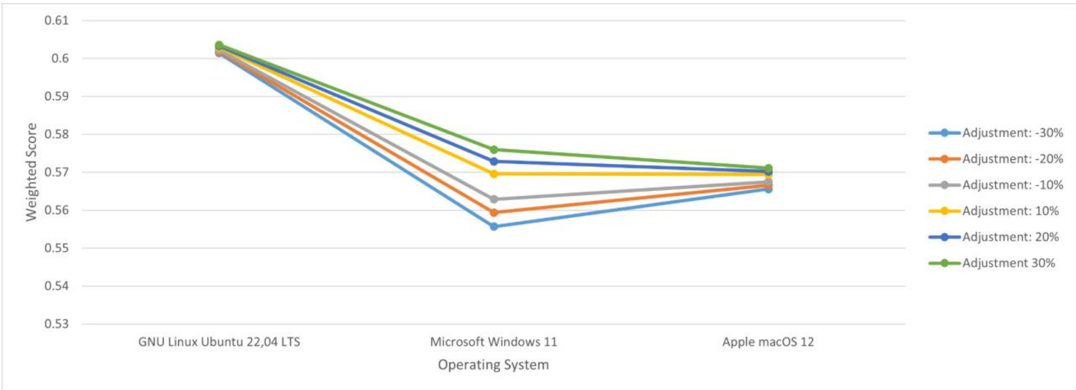

Figure 5. Sensitivity analysis for implementation.

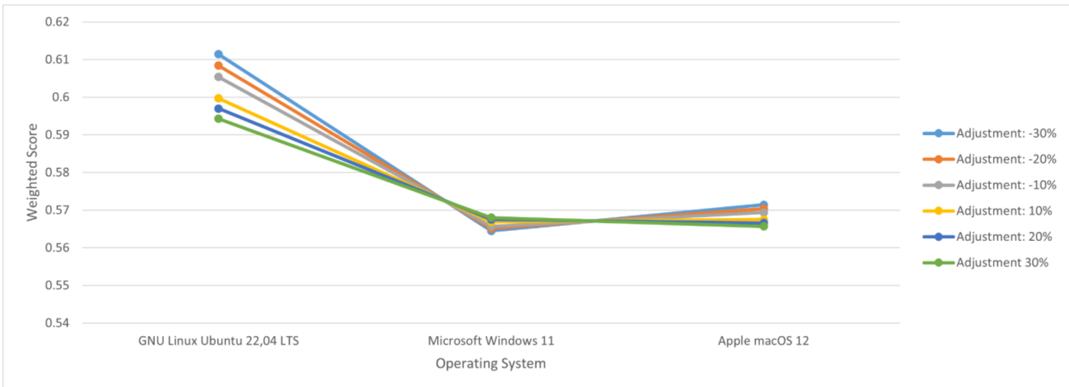

Figure 6. Sensitivity analysis for support.

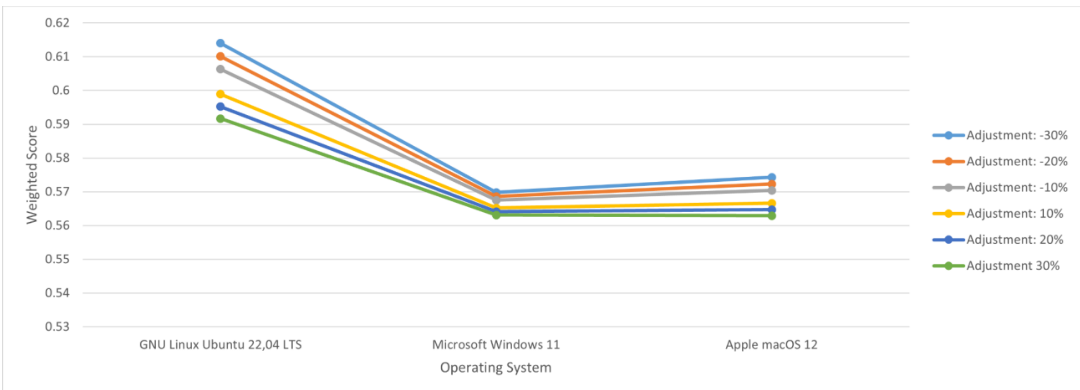

Figure 7. Sensitivity analysis for documentation.

Key observations from the sensitivity analysis:

- Performance (C1): Increasing the weight of Performance by 30% slightly enhanced the score of GNU Linux Ubuntu 22.04 LTS, while Microsoft Windows 11 and Apple macOS 12 experienced minimal variations. The overall ranking remained consistent.
- Cost (C2): Adjustments to Cost showed marginal changes in the scores of all three systems. GNU Linux Ubuntu 22.04 LTS remained the highest-ranked option, demonstrating the robustness of the proposed model in cost-sensitive scenarios.
- Security (C3): Variations in the Security weights had a more pronounced effect on the scores, but the overall ranking of the alternatives remained stable, highlighting the adaptability of the model to security-related considerations.
- Other Criteria (C4–C7): Adjustments to criteria such as Usability, Implementation, Support, and Documentation introduced minor fluctuations in the scores but did not alter the final ranking, further confirming the reliability of the framework.

The sensitivity analysis confirms that the proposed decision-making model is robust and reliable, as the rankings of the operating systems remained consistent across a range of weight adjustments for all criteria. This reinforces the practical applicability of the model for decision-making in educational environments.

6. Discussion

The results obtained in this study provide significant insights into the selection of operating systems for educational institutions. For instance, GNU Linux Ubuntu 22.04 LTS emerged as the highest-ranked operating system due to its cost-effectiveness and adaptability to resource-constrained environments. This finding highlights its potential for implementation in public schools or institutions with limited budgets. On the other hand, Microsoft Windows 11 and Apple macOS 12 showed strengths in usability and support, making them more suitable for institutions prioritizing user experience or requiring extensive technical support.

These results suggest that the proposed model can serve as a practical decision-making tool for educational administrators, helping them align their technological choices with institutional priorities. By applying this framework, institutions can systematically evaluate their options, minimize resource wastage, and optimize their IT infrastructure to support teaching and learning activities.

This paper explored the application of a novel decision-making approach in education through multi-criteria decision-making (MCDM) for selecting the appropriate OS. By using the PIPRECIA method, which integrates the assessment of the weight criteria with group decision-making and utility analysis (NWA), a balance can be achieved between the needs of the educational institution and the expertise of the IT professionals.

One of the key aspects of this study was considering the complex challenges faced by decision-makers in education when choosing an OS. Traditional approaches often rely on the subjective opinions of individuals, which can lead to biased decisions and the incomplete consideration of all relevant factors. In this context, the PIPRECIA method, in combination with the utility analysis (NWA), represents an innovative approach that enables decision-makers to make decisions based on objective criteria and group consensus.

It is important to emphasize that educational institutions have specific needs and goals that often differ from other sectors. Therefore, the process of selecting an OS for educational purposes requires a special approach that takes into account the academic, technical, and strategic priorities. The application of these methods provides flexibility in defining weight criteria and allows for adaptation to the specific needs of each educational institution.

Furthermore, it is important to highlight the potential broader application of this approach in education. While this paper focused on the selection of an OS, such an approach can be applied to other decisions in education such as choosing teaching materials, project planning, or resource management. This opens up the possibility of improving the efficiency and effectiveness of decision-making in education.

Considering all of the factors above-mentioned, we can conclude that the application of this method for selecting an OS in education brings numerous benefits. This method allows for an objective consideration of all relevant factors, reduces subjectivity in decision-making, and enables group decision-making that integrates different perspectives and expertise. Through a well-guided process of selecting an OS, educational institutions can improve their IT infrastructure, support teaching activities, and enhance the overall experience of students and staff.

Finally, the ranking of the proposed OS from our analysis can be clearly seen next to Figure 8, along with the minimal differences shown, but again, tailored to each institution depending on their specifications.

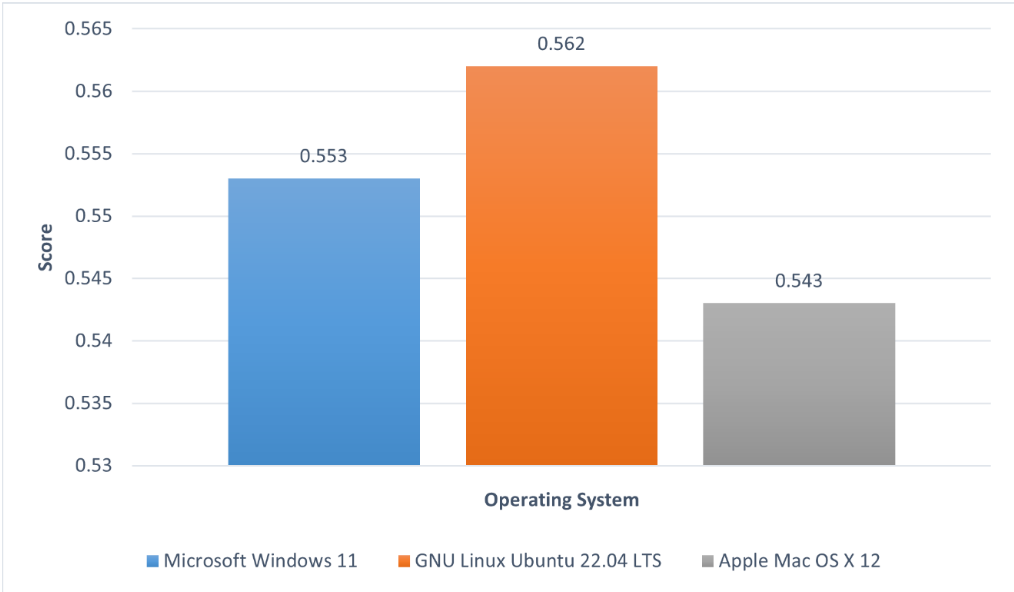

Figure 8. Display of the final results.

While the PIPRECIA-NWA model is effective for decisions involving a limited number of criteria and alternatives, its simplicity may not be suitable for highly complex decision-making scenarios that require hierarchical analysis or a larger pool of alternatives. In such cases, methodologies like AHP might offer greater robustness. However, for decisions like selecting an OS in education, where time and resource constraints are significant, the simplicity of the PIPRECIA-NWA approach becomes a distinct advantage.

Despite its contributions, the proposed model has several limitations that should be acknowledged. Firstly, the model relies on predefined criteria and weights, which, although carefully selected, may not fully capture the unique needs of all educational institutions. Customization of criteria for specific contexts would require additional effort and expertise.

Secondly, the model assumes that decision-makers have sufficient knowledge and access to accurate data for evaluating alternatives. In practice, this may not always be the case, particularly in smaller institutions with limited access to IT expertise.

Thirdly, while the sensitivity analysis demonstrated the robustness of the model, its scope was limited to variations in criteria weights. Future studies could explore more comprehensive scenarios including dynamic changes in institutional priorities or unexpected constraints.

Finally, the study focused on three specific operating systems. Expanding the scope to include additional options or emerging technologies would provide a more comprehensive evaluation framework. Addressing these limitations in future research would further enhance the model's applicability and relevance.

6.1. Limitations of the Study

This study was subject to several limitations that should be acknowledged. First, the analysis was conducted using a limited number of criteria and alternatives, which may not have fully captured the complexities of operating system selection in diverse educational settings. Second, the study relied on the subjective judgments of a small group of experts, which, although carefully chosen for their expertise, could have introduced bias into the weighting and ranking processes. Third, the proposed model was specifically tailored for the educational context, and its applicability to other domains may require further adjustments and validation.

6.2. Suggestions for Future Work

Future research could address these limitations by expanding the number of criteria and alternatives included in the analysis to enhance the robustness of the model. Additionally, integrating advanced techniques, such as fuzzy logic, could improve the representativeness and flexibility of the evaluations, particularly in cases with higher levels of uncertainty. Finally, applying the model in different domains, such as healthcare or corporate IT environments, could validate its broader applicability and identify potential improvements.

7. Conclusions

Decision-making represents an essential part of everyday operations and management, both in educational institutions and other sectors. Regardless of the context, the decision-making process is often complex and requires the careful consideration of various factors, criteria, and potential consequences. When faced with multi-criteria decision-making, the situation becomes even more challenging, as it requires taking into account the different perspectives, priorities, and expertise of the involved parties.

In this study, we focused on developing a model that enables efficient decision-making in multi-criteria environments, using group decision-making involving not only multiple decision-makers, but also IT experts. This approach represents a step forward in the decision-making process, providing structure, a systematic approach, and balance among the different parties involved.

One of the key advantages of this model is the elimination of bias in decision-making. Through group decision-making, multiple decision-makers have the opportunity to express their views, priorities, and opinions, while simultaneously considering the expertise of IT professionals. This allows for a more comprehensive understanding of the situation and the making of informed decisions that take into account different perspectives and expertise.

Moreover, the application of a mathematical model in the decision-making process provides additional security and reliability. Through data analysis and quantitative evaluation of alternatives, decision-makers have a clearer picture of the potential consequences of each decision, contributing to a better understanding of the situation and risk reduction.

It is important to note that this model not only facilitates decision-making, but also promotes collaboration and teamwork among participants. Through a joint decision-making process, decision-makers have the opportunity to exchange ideas, consider different options, and collectively find the best solution for the given situation. This is not only useful for achieving better results, but also for building team spirit and mutual trust.

While we have highlighted numerous advantages of this model, it is also important to recognize some of its limitations. For example, the complexity of the process may require additional time and resources, which can be challenging in situations with limited time or budget. Additionally, dependence on available information and expertise may limit the applicability of the model in some specific situations.

In conclusion, the multi-criteria decision-making model that we developed is a useful tool for efficient decision-making in complex situations. Through a combination of group decision-making, a mathematical model, and the IT experts' expertise, this model provides structure, balance, and reliability in the decision-making process. Overall, these findings contribute to improving the decision-making process and promoting effective management in various organizational contexts.

This study highlights the efficiency of the PIPRECIA-NWA model in providing valid and reliable decisions within specific contexts. The findings of this study are particularly valuable for educational administrators, IT managers, and policymakers tasked with making informed decisions about operating systems in educational environments. By providing a systematic and transparent decision-making framework, the proposed PIPRECIA-NWA model equips decision-makers with the tools to balance technical, financial, and usability criteria effectively. Additionally, the model can serve as a reference for researchers and practitioners in other fields who are interested in applying multi-criteria decision-making methods to similar challenges. The model is particularly effective in scenarios where simplicity and speed are prioritized such as in educational institutions. By offering a clear and transparent framework, the model enables decision-makers to achieve high-quality results without the need for extensive resources or time.

While the PIPRECIA-NWA model demonstrated its effectiveness in providing a structured and transparent decision-making framework, its applicability is primarily suited to scenarios with a limited number of criteria and alternatives. For instance, the model is particularly useful in educational institutions with constrained resources, where decision-making needs to be both efficient and objective. Its simplicity and ease of implementation make it an ideal choice for such environments, ensuring that decision-makers can achieve reliable outcomes without extensive computational or time demands.

However, the model may be less suitable for highly complex decision-making scenarios involving a large number of alternatives or where significant uncertainty exists in the evaluation criteria. In such cases, integrating fuzzy logic or hybrid decision-making approaches could provide greater flexibility and adaptability.

To maximize the effectiveness of the PIPRECIA-NWA model, the following guidelines are recommended:

- Use in resource-constrained settings: The model is ideal for institutions seeking cost-effective and time-efficient decision-making solutions.
- Focus on clear and measurable criteria: The simplicity of the model relies on well-defined criteria and reliable data for evaluation.
- Tailor criteria to context: While the predefined criteria used in this study are relevant to educational settings, decision-makers are encouraged to adapt the criteria to address the specific needs of their institutions.
- Consider future extensions: For more complex scenarios, decision-makers should explore integrating the model with advanced methodologies, such as fuzzy logic, to better handle uncertainty and larger datasets.

By adhering to these guidelines, decision-makers can effectively apply the PIPRECIA-NWA model in a variety of educational contexts, ensuring that the decision-making process is both robust and aligned with institutional priorities.

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